Daniel Mitroshkov

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EDUCATION

University of Washington Seattle, WA
Bachelor of Science in Computer Science; Minor in Philosophy. *Expected June 2028*
Relevant Coursework: Human/AI Interaction, Deep Learning, Modern Concurrency, Systems Programming, Data Structures and Algorithms, Hardware & Software Interface, Linear Algebra, Probability and Statistics in Computing

SKILLS

Programming: Python, JavaScript, TypeScript, HTML/CSS, Rust, C++, Java, SQL, C#
AI & Data: PyTorch, LangChain, pandas, DuckDB, Jupyter
Engineering & Cloud: AWS (Lambda, S3, EventBridge, ECR, CloudFront), Docker, Git, pytest, JUnit, Node.js

EXPERIENCE

Undergraduate Researcher / REU Intern December 2025 – Present
Interactive Data Lab *Seattle, WA*
- Developing ADIEU, an LLM-based extension of prior algorithmic simulation research, adding agent simulation and evaluation capabilities
- Reviewed 50+ AI/HCI papers to inform feature development and evaluated 5+ LLM-powered data-analysis workflows
- Building modular research software and data-processing tools with Python and DuckDB to support reproducible experiments
- Collaborating with graduate researchers on software design, debugging, validation, and code review

Teaching Assistant September 2025 – Present
Paul G. Allen School of Computer Science and Engineering *Seattle, WA*
- Coordinating with CSE 394 course staff to help students begin conducting computer science research
- Supported 25+ students quarterly in CSE 197Y, a workshop for UW's introductory programming series, leading 3 hrs/week of one-on-one tutoring in programming fundamentals and problem solving
- Developed 80+ JUnit test cases, refined course materials, and co-led staff meetings to align grading and support

IT Technician Intern June 2022 – July 2024
Richland School District *Richland, WA*
- Resolved 500+ help-desk tickets across 21 schools while supporting a fleet of 6,000+ devices
- Improved Excel and Google Sheets inventory tools tracking 200+ shipments

PROJECTS

Empathetic Help-Seeking System April 2026 – Present
Human/AI Interaction | UW
- Built an LLM-assisted response-rewriting prototype and compared 15+ strategies across 3 models
- Conducted and analyzed a 20-participant study measuring perceived empathy and willingness to seek help; found the system improved help-seeking even among participants resistant to AI, by surfacing perspectives and suggestions they would not have considered independently

The Harrowing Tale of Human Trafficking Winter 2026
cse442.pages.cs.washington.edu/26wi/fp/the-harrowing-tale-of-human-trafficking
CSE 442: Data Visualization | UW
- Co-created an interactive exploration of reported human trafficking across five global regions using CTDC's privacy-preserving synthetic dataset
- Paired a scroll-driven 3D globe with linked demographic charts, country-level exploration, and age and gender filters, combining guided storytelling with open-ended analysis while making reporting biases and data limitations explicit
- Earned a perfect score, a rare distinction in CSE 442

contrailλ May 2026
AWS Hacks, 4th Place
- Built an event-driven serverless pipeline across 5 AWS services in 24 hours, transforming real-time NOAA GFS data into global forecasts of contrail-favorable airspace
- Implemented atmospheric scoring on global 0.25 degree grids, producing ~1,000 polygons across 5 intensity tiers every 6 hours
- Containerized Python inference with Docker; achieved ~20s runtime per scene and autonomous operation under $1/day

Imaginarium October 2025
DubHacks '25
- Built a real-time AI story-generation system with context-aware responses to user input in under 24 hours
- Implemented authentication, data persistence, and payment integration to support premium features

Snap Select August 2025 – Present
- Developed an OCR screenshot tool with clipboard integration, achieving ~30ms end-to-end latency
- Achieved 94% accuracy scanning text from image types of varying quality